

Reviewer Pack — Minimal Local Index and Frege Proof Depth Correlation for k...

Entry be7b7e974a78 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Index and Frege Proof Depth Correlation for k-CNF

Entry ID: be7b7e974a78 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 05:55:53 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Local Systems) **Field B** (complexity object): Boolean Satisfiability (Frege Proof Complexity)

Statement:

For every k-CNF instance with n variables, the minimal local index of its induced simplicial complex is linearly correlated with its Frege proof depth such that $\text{LocalIndex}(C) = \Theta(\text{FregeProofDepth}(C))$ for any k-CNF formula C .

Rationale (proposer's reasoning):

The study of local systems in algebraic topology can provide insights into the structure of Boolean satisfiability problems. Minimal local index captures the complexity of the induced simplicial complex, which might reflect the structural depth of Frege proofs for SAT instances.

Taxonomy category: local_systems_frege_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1561304bd42a91ca

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across 30 random seed generations, the correlation coefficient between the computed minimal local index and Frege proof depth is ≥ 0.7 and no seed instance has a local index > 10 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'algebraic topology' AND 'Frege proof complexity' AND 'k-CNF' - 'local index' AND 'Frege proof depth' AND 'k-CNF' algorithm - 'induced simplicial complex' AND 'Boolean satisfiability' AND 'linear correlation'

Top relevant hits considered: - [http://arxiv.org/abs/1002.3702v2] Algebraic K-theory over the infinite dihedral group: a controlled topology approach - [http://arxiv.org/abs/1108.4976v2] Cyclic A_∞ Structures and Deligne's Conjecture - [http://arxiv.org/abs/1609.07316v2] Equivariant cohomology of cohomogeneity one actions: the topological case - [http://arxiv.org/abs/1208.1862v2] A transformation rule for the index of commuting operators - [http://arxiv.org/abs/math/0509113v2] A Proof of The Fundamental Theorem of Algebra - [http://arxiv.org/abs/0805.1411v3] Cyclic cocycles on deformation quantizations and higher index theorems - [http://arxiv.org/abs/0910.1498v3] Dualizing complex of the face ring of a simplicial poset - [http://arxiv.org/abs/1506.05567v2] Integral Foliated Simplicial Volume of Aspherical Manifolds

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        max_row = i
        for j in range(i+1, m):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]
        for j in range(i+1, m):
            factor = A[j][i] / A[i][i]
            for k in range(n):
                A[j][k] -= factor * A[i][k]
    return A

def rank(A):
    rref = gaussian_elimination(A)
    rank = 0
    for row in rref:
        if any(row):
            rank += 1
    return rank
```

```

def random_k_cnf(n, m):
    variables = list(range(1, n+1))
    clauses = set()
    while len(clauses) < m:
        clause = random.sample(variables, random.randint(2, min(m, n)))
        if all(v not in c for v in clause for c in clauses):
            clauses.add(tuple(sorted(clause)))
    return clauses

def frege_proof_depth(k_cnf):
    # Placeholder function for Frege proof depth calculation
    # This is a dummy implementation and should be replaced with the actual algorithm
    return len(k_cnf) * 2

def minimal_local_index(simplicial_complex):
    # Placeholder function for minimal local index calculation
    # This is a dummy implementation and should be replaced with the actual algorithm
    return rank(simplicial_complex)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 40
    m = n * (n - 1) // 2
    k_cnf = random_k_cnf(n, m)
    simplicial_complex = [[], [], []] + [list(range(1, n+1))] * (n-3)
    local_index = minimal_local_index(simplicial_complex)
    frege_depth = frege_proof_depth(k_cnf)

    if local_index > 10:
        return {
            "metric_name": "LocalIndex",
            "metric_value": local_index,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": "local_index_too_high"
        }

    return {
        "metric_name": "Correlation",
        "metric_value": local_index * frege_depth, # Dummy correlation for testing
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")

```

```

results.append(trial_result)

mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='local_index_too_high' first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the pre-registered support condition could not be unambiguously met. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12477
2	preregistration	ollama_remote	glm4:latest	0	0	9355
3	novelty	ollama_remote	glm4:latest	0	0	8739
4	novelty	ollama_remote	glm4:latest	0	0	10383
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19002
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13855
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11824
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11060
9	judge	ollama_remote	glm4:latest	0	0	13181

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 109873 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/be7b7e974a78.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/be7b7e974a78.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/be7b7e974a78.tar.gz (if generated)